Physics 5A Discussion (Week 1)

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Example: Eric is working on an experiment in his Physics 5AL lab. He saw a ping-pong ball was initially moving to the right at a constant speed 1 m/s for 2 seconds. Then the ball hit the wall and moved back to the left at a constant speed 0.5 m/s for another 5 seconds. For now we ignore any friction.

In this case, we can set up the coordinate system with x-axis pointing to the right as usual. For the first two seconds, the ball moved to the right, so it had velocity $v_\mathrm{i} = 1$ m/s. After the ball hit the wall, it moved to the left, so its velocity became $v_\mathrm{f} = -0.5$ m/s (the negative sign indicates the direction to the left). The total <u>distance</u> it has traveled (think about the mileage reading on a used car) is

$$|\Delta x| = |v_\mathrm{i}| \cdot (2\ \mathrm{s}) + |v_\mathrm{f}| \cdot (5\ \mathrm{s}) = (1\ \mathrm{m/s}) \cdot (2\ \mathrm{s}) + (0.5\ \mathrm{m/s}) \cdot (5\ \mathrm{s}) = 4.5\ \mathrm{m}\,.$$

But the <u>displacement</u> it has traveled is calculated differently, where we need to consider the "net change" of its position in this case,

$$\Delta x = v_\mathrm{i} \cdot (2\ \mathrm{s}) + v_\mathrm{f} \cdot (5\ \mathrm{s}) = (1\ \mathrm{m/s}) \cdot (2\ \mathrm{s}) + (-0.5\ \mathrm{m/s}) \cdot (5\ \mathrm{s}) = -0.5\ \mathrm{m}\,.$$

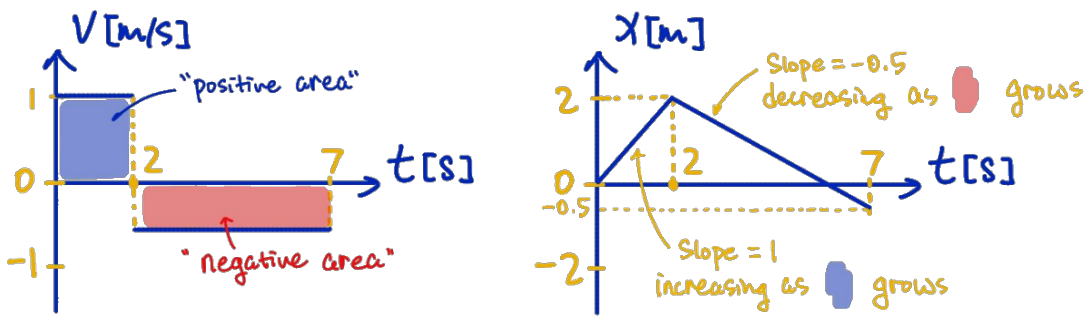
This is because it ended at a position 0.5 meters to the <u>left</u> of where it started. Next we calculate the <u>average speed</u> over the first 7 seconds,

$$\overline{s} = \frac{|\Delta x|}{7\ \mathrm{s}} = \frac{4.5\ \mathrm{m/s}}{7\ \mathrm{s}} = 0.643\ \mathrm{m/s}\,.$$

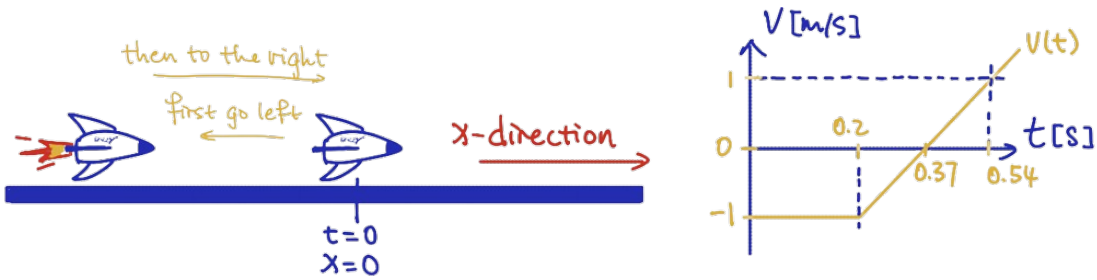
We don't care about the direction the ball was traveling when we calculate the average speed, but we do when we calculate its <u>average velocity</u>:

$$\overline{v} = \frac{\Delta x}{7\ \mathrm{s}} = \frac{-0.5\ \mathrm{m/s}}{7\ \mathrm{s}} = -0.071\,\mathrm{m/s}\,.$$

The acceleration of the ball in the first 2 seconds is 0 m/s^2, and also 0 m/s^2 after the ball has hit the wall. But at the instant the ball hit the wall, it was "pushed" (hence accelerated) to the left by the wall for a very short amount of time but with a very large magnitude, thus the velocity (and direction of motion) was changed. If we were to draw the velocity/position vs. time plot for this ball, we would have the followings.



Notice that the "area" under the velocity curve (the shaded region bounded between the x-axis and the velocity curve, from $t = 0$ to any other time) corresponds to the value of the position curve. The slope of the position curve corresponds to the value of the velocity curve.



Problem 1: A rocket is mounted on a track shown in the figure. The rocket initially has a speed moving to the left, then after a while the boost on the rocket is switched on, providing a constant acceleration to the right. Jinyan has obtained a velocity vs. time graph for the rocket as shown, and he asks you to draw the position vs. time, and acceleration vs. time graph for the rocket (You can assume that the position of the rocket at $t = 0$ s is $x = 0$ m). Remember to label the axes.

(Hint: This problem is easy if you know calculus, but for Physics 5A, we guarantee that you can solve all problems without calculus. Think about "area under the curve" for finding the position plot, and "slope of the curve" for finding the acceleration plot.)

Problem 2: Calculate the displacement, traveled distance, average speed, and average velocity of the rocket in the first 0.54 seconds.

Problem 3: During the first 0.54 seconds, find the time instances that the rocket has (a) its highest speed, (b) highest velocity, (c) greatest displacement, (d) longest distance traveled.

Problem 1 (solution): See the end of this file :)

Problem 2 (solution): First we find the displacement, which is the "net change" of the rocket's position in the first 0.54 seconds. This only involves two quantities, final position x_f and initial position x_i of the rocket. At $t = 0.54$ seconds, we have $x_\mathrm{f} = -0.2$ meters (obtained from the position graph). At $t = 0$ second, we have $x_\mathrm{i} = 0$ meter. Then we can write

$$\text{Displacement} = x_\mathrm{f} - x_\mathrm{i} = (-0.2 - 0)\,\mathrm{m} = -0.2\,\mathrm{m}$$

The traveled distance is like the mileage reading in your car, so the more time it spends running on the road, the longer the distance traveled. In this problem, the rocket traveled to the left for 0.2 meters, continue to the left for 0.085 meters, then to the right for 0.085 meters, we add these three distances without negative sign,

$$\text{Distance} = (0.2 + 0.085 + 0.085)\,\mathrm{m} = 0.37\,\mathrm{m}$$

Having displacement and distance calculated, average speed and average velocity can be calculated easily from their definitions. For average speed,

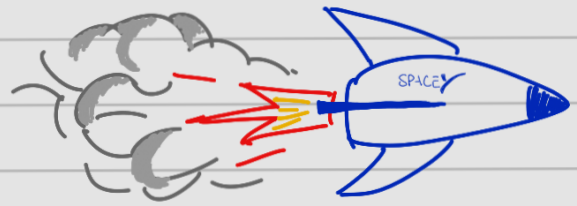
$$\overline{s} = \frac{\text{Distance}}{\Delta t} = \frac{0.37\,\mathrm{m}}{0.54\,\mathrm{s}} = 0.685\,\mathrm{m/s}\,.$$

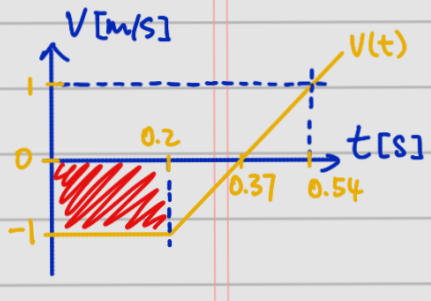
For average velocity

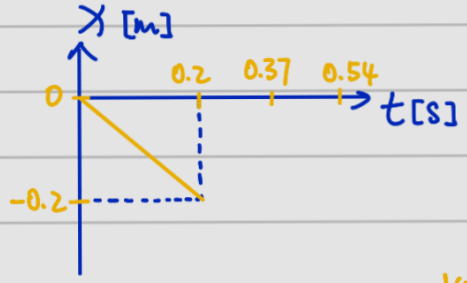
$$\overline{v} = \frac{\text{Displacement}}{\Delta t} = \frac{-0.2\,\mathrm{m}}{0.54\,\mathrm{s}} = 0.370\,\mathrm{m/s}\,.$$

Problem 3 (solution): The speed of the rocket is the absolute value of its velocity, so we look at the velocity graph, we see that the rocket has speed 1 m/s for the first 0.2 seconds, and also has speed 1 m/s at $t-0.54$ s. At all other time, the rocket has speed less than 1 m/s. <u>(a) The rocket attains its highest speed at time between 0 and 0.2 seconds, and also at $t = 0.54$ seconds</u>. For highest velocity, we only look at the velocity plot and find its highest value, which is <u>(b) $v = 1$ m/s at time $t = 0.54$ second</u>. Usually "highest velocity" is not a very useful, or informative, quantity because it does not account for the fact that the rocket can move to the left (negative velocity) at a very high speed, but it signifies highest speed towards the direction "you want" (the positive direction). Since the displacement of the rocket has a plus/minus sign, and based on the position graph, the displacement $\Delta x = x_\mathrm{f} - x_\mathrm{i} = x_\mathrm{f}$ (because the rocket starts from x_i at time 0) is always negative, so the <u>(c) largest number of displacement is $\Delta x = 0$ at time $t = 0$</u>. This is a tricky question, sometimes you might also be asked for the longest distance away from the origin, in which case is at time 0.38, when the rocket is farthest to the left from where it started. The longest distance that the rocket has traveled (not the longest distance from the origin) has been calculated in problem 2, as the "distance traveled" is like the mileage reading in your car, <u>(d) the greatest number of that is at time $t = 0.54$ s</u>.

Problem 1







to find displacement from $t=0$ to $t=0.2$ seconds $\quad V=\frac{\Delta x}{\Delta t} \quad \Delta x = V\cdot\Delta t \quad V=-1\ m/s \quad \Delta t = 0.2-0=0.2\ s$

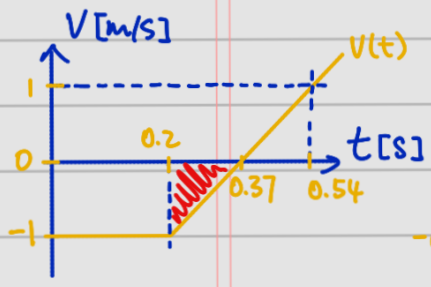
$\hookrightarrow \Delta x = -0.2\ m$

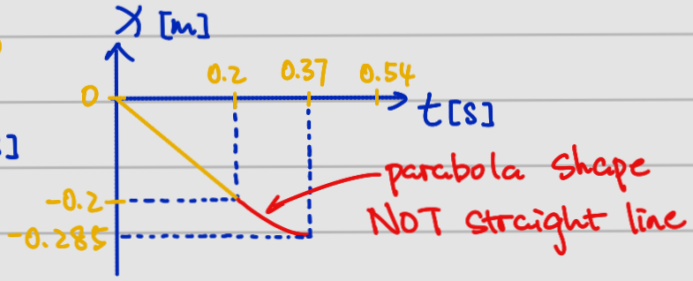
so at $t=0.2s$ $\quad x = x_i + \Delta x = 0m - 0.2m = -0.2m$ $\quad$ this is exactly the "area" of ▬ up to negative sign

connect $\begin{Bmatrix} t=0 \\ x=0 \end{Bmatrix}$ with $\begin{Bmatrix} t=0.2 \\ x=-0.2 \end{Bmatrix}$ using a straight line because:

① velocity is constant "$V=\frac{\Delta x}{\Delta t}$" so the increase/decrease in x per unit time is the same

② "the car is moving at a constant speed" → distance changes steadily





to find displacement from $t=0.2$ to $t=0.37$ seconds

velocity is no longer constant, but with similar reasoning, use "area" to find Δx

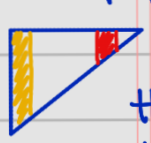
$\Delta x =$ "area" of ▼ $= \frac{1}{2}(0.37-0.2)s \cdot (-1\ m/s) = -0.085\ m$

$\Rightarrow x_{t=0.37s} = x_{t=0.2s} + \Delta x = -0.2 - 0.085 = -0.285\ m$

What is happening at this stage?

① from velocity graph, v starts to change at $t=0.2\ s$, so boost is turned on at $t=0.2s$. But ⊳ was initially moving ←, so the boost will first slow it down, still moving left slowly, until it stops. So x decreases but with a smaller and smaller rate

② With physical intuition in ①, mathematically.



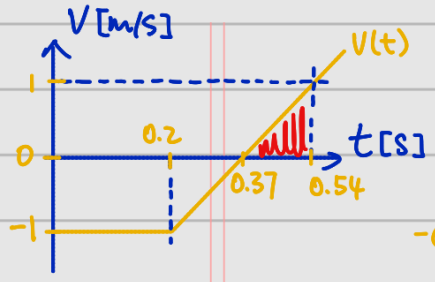
▨'s area is larger than that of ▬

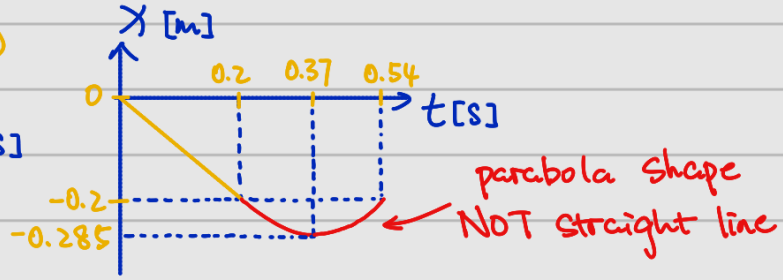
these "areas" are the change in position of ⊳

↳ x decreases by a lot at the beginning, the change in position is smaller and smaller → a parabola shape

③ $V=0\ m/s$ at $t=0.37s$ ⇒ the ⊳ stops at $t=0.37s$ → bottom of the parabola

↳ slope of position curve $= 0 = V$ at $t=0.37s$.





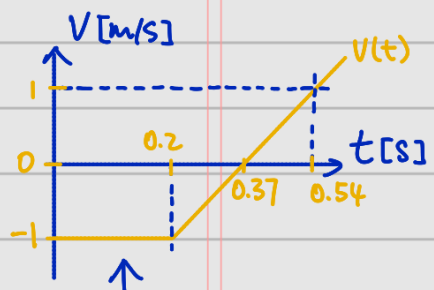
Same as what we did above for the "negative triangle"
from $t=0.37$ to $t=0.54\,s$ $\quad \Delta x =$ "area" of ◭ $= \frac{1}{2}(0.37-0.2)s \cdot (1\ m/s) = 0.085\ m$

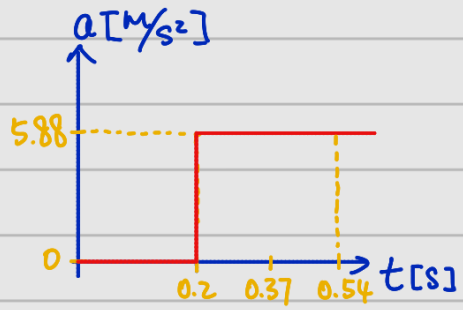
two triangles have the same area up to a negative sign. What does this mean?

① Boost is turned on at $t=0.2s$. first slows down the leftward motion. then the rocket comes to a stop, then makes the rocket go faster to the right.

② When slowing down. change in position is smaller and smaller
When speeding up. change in position is larger and larger.
→ parabola shape for position curve

③ Two triangles have same size ⇒ go back to $x=0.2\ m$





from $t=0$ to $t=0.2\,s$
velocity did not change

$a = \frac{\Delta v}{\Delta t} = 0\ m/s^2$ $\quad \left(\text{Slope} = \frac{\Delta y}{\Delta x}\right.$ ← Δy is Δv in our context, ← Δx is Δt in our context$\left.\right)$

So $a=0\ m/s^2$ from $t=0$ to $t=0.2s$ flat line Boost is off

from $t=0.2$ to $t=0.54\,s$
velocity change in a constant rate

$$a = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i} = \frac{1-(-1)}{0.54-0.2}\ m/s^2 = 5.88\ m/s^2$$

So $a=5.88\ m/s^2$ from $t=0.2$ to $t=0.54\,s$. flat line Boost is on
making velocity increase

think about a object moving left but you are trying to push it to the right